Predicting Student Exam Scores from Background Features Using Linear Regression and Random Forest

1st **Sujithra R**
Department of Computer Science and Engineering
KPR Institute of Engineering and Technology
Coimbatore, India

2nd **Thabu Prabaa A B**
Department of Computer Science and Engineering
KPR Institute of Engineering and Technology
Coimbatore, India

3rd **Visalini S T**
Department of Computer Science and Engineering
KPR Institute of Engineering and Technology
Coimbatore, India

4th **Santhini A**
Department of Computer Science and Engineering
KPR Institute of Engineering and Technology Coimbatore, India

5th **Pradeepa N**
Department of Computer Science and Engineering
KPR Institute of Engineering and Technology Coimbatore, India

***Abstract*—Early estimates of student performance help teachers plan support before final results are known. This work predicts the average of math, reading and writing scores of 1,000 students from five background features: gender, race/ethnicity, parental level of education, lunch type and test preparation course. The dataset has no missing values or duplicate records. The three scores were averaged to form the target and were left out of the inputs to avoid leakage. Nominal features were one-hot encoded and parental education was ordinal-encoded. Linear Regression and Random Forest Regressor were trained on an 80:20 split and evaluated with R^2, MAE, RMSE and five-fold cross-validated R^2. Linear Regression performed best, with a test R^2 of 0.178, an MAE of 10.39 and an RMSE of 13.28 score points, against 0.127, 10.67 and 13.68 for Random Forest. Lunch type, test preparation and parental education carried the highest feature importance. The low R^2 values show that these background features explain only a small part of the variation in scores. The results give a reproducible baseline for later work that adds study-behaviour data.**

***Keywords*—student performance prediction, regression, educational data mining, linear regression, random forest**

I. INTRODUCTION

Exam results shape a student's progress, course choices and access to extra support. Teachers usually see these results only after the assessment is over. A model that estimates performance from information already on record could point to students who may need help sooner. Educational data mining studies this kind of problem by applying data mining methods to educational data [1].

Machine learning is now widely used for this task. Early work on Portuguese secondary schools showed that earlier grades are strong predictors of later results [2]. Later studies predicted master's performance from records that are easy to collect at enrolment [3], examined the factors behind academic probation [4], and combined random forests with a genetic algorithm for multi-class prediction [5]. Regression has also been used to predict grades directly, with linear models [6], [7] and with synthetic data generation [8].

Two limitations stand out in this literature. First, many datasets are small and come from one institution or course, so findings may not transfer [9], [10]. Second, strong results often depend on earlier grades or behaviour records [2], [3], which may not exist when a prediction is most useful. It is less clear how much of a student's score can be predicted from background information alone.

This work uses the Students Performance in Exams dataset, a public collection of 1,000 records [11]. Each record has five background features (gender, race/ethnicity, parental level of education, lunch type and test preparation course) and three subject scores in math, reading and writing.

In this work, the three scores are averaged into one target, and the models receive only the five background features. Linear Regression serves as an interpretable baseline [12], and Random Forest as a nonlinear ensemble [13]. Both are built with scikit-learn [14] and compared using R^2, MAE and RMSE [15].

The main contributions are a leakage-free setup in which no score column is used as an input, a comparison of the two models on a held-out split and with five-fold cross-validation, and an analysis of which background features carry the most information, using chi-squared scores and Random Forest importances.

II. LITERATURE REVIEW

Table I compares eight recent studies. Regression-based work is common. Santiketa et al. [6] predicted the GPA of 853 high-school students, and Linear Regression reached an R^2 of 0.8911 using earlier scores. Asthana et al. [7] compared five regression models and found Linear Regression the most accurate, at 97%. Ying and Ma [8] reported that regression beat classification on a five-level grading scheme and that synthetic data raised accuracy for SVM, neural network and Random Forest models. Sayed et al. [10] found the same advantage for regression on 922 postgraduate records, but their best R^2 was only 0.383.

Other studies focus on tree ensembles and interpretation. Chen and Liu [5] combined Random Forest with a genetic algorithm and reported an average accuracy of 93.11%. Wang and Luo [9] used XGBoost with SHAP values on 87 students to explain which factors drive achievement. Studies of feature influence found that semester grades led the ranking for master's students [3], while study duration and secondaryschool performance led for students on probation [4].

Three points follow from Table I. Linear models are often competitive with more complex ones [6], [7]. The highest accuracies tend to rely on earlier grades or learning data, and

the one study with a moderate R^2 used mixed academic and demographic records [10]. Most datasets are small or come from a single site. The present work differs by using only background features from a public dataset and by reporting errors in score points as well as R^2.

III. PROPOSED SYSTEM

As shown in Fig. 1, the work follows a sequence of data inspection, target creation, feature encoding, model training and evaluation. The steps below match the Colab notebook.

TABLE I
COMPARISON OF RELATED WORK

Year	Ref.	Dataset/Application	Method	Main Finding	Limitation
2025	[10]	922 postgraduate public health records, Egypt	Classification and regression models; ensemble	Regression beat classification; softvoting ensemble: accuracy 74.25%, R^2 0.383	Single institution; moderate R^2; no psychosocial data
2024	[6]	853 high-school students; GPA and grade prediction	Linear Regression, SVM, KNN, ANN, Random Forest	Linear Regression best for GPA (R^2 0.8911); RF and KNN 100% grade accuracy	One school programme; relies on past grades
2024	[9]	87 students, one university course, China	XGBoost with SHAP	Interpretable prediction; shows effect of instructional mode	Very small sample; one course
2024	[5]	Student performance case study; multi-class	Random Forest with genetic algorithm; IG and Laplacian score selection	Average accuracy 93.11%, at least 2.25% above baselines	Single case study; added tuning complexity
2024	[8]	Five-level grade prediction; small datasets	SVM, NN, RF with regression and synthetic data	Regression beat classification; accuracy up to +21.9%, +15.6%, +6.6% (SVM, NN, RF)	Gains depend on quality of generated data
2023	[7]	Students with 'learning coefficients' as features	DT, RF, SVR, Linear Regression, ANN	Linear Regression most accurate (97%)	Needs adaptive-assessment data to derive coefficients
2023	[4]	College students on academic probation, Oman	InfoGain; Logit Boost, Vote, Bagging	Study duration and secondary-school performance ranked top	Single institution; probation cases only
2022	[3]	700+ master's records, University of Mannheim	Data mining algorithms with feature importance	Semester grades most important, then distance and culture	One faculty; larger data and other programmes needed

A. Data Acquisition

The notebook loads StudentsPerformance.csv with pandas. The file has 1,000 rows and eight columns: five categorical features and three integer scores. The mean scores are 66.09 (math), 69.17 (reading) and 68.05 (writing), with standard deviations between 14.6 and 15.2. The lowest values are 0 for math, 17 for reading and 10 for writing.

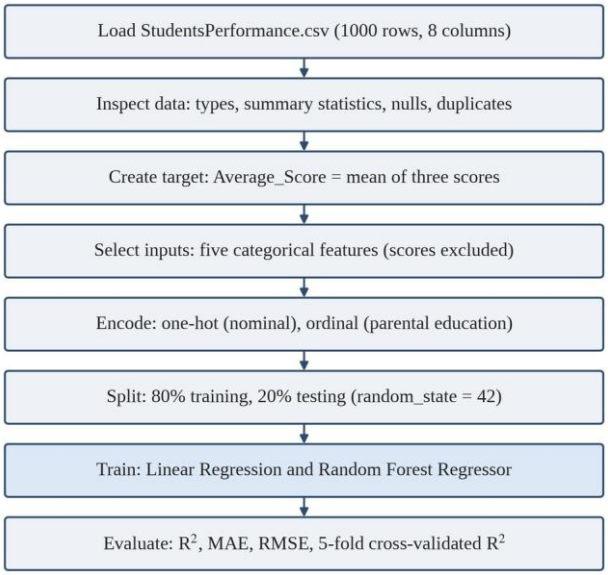


Fig. 1. Overall workflow of the proposed work.

B. Data Preprocessing

Basic checks found no missing values and no duplicate records, so no cleaning was needed. Total_Score and Average_Score were then derived from the three scores. Each score correlates strongly with Average_Score (0.92 to 0.97), which is why the scores are kept out of the inputs. Grouping the records by the five categorical features gives only 211 distinct combinations. The notebook also demonstrated label encoding, standard scaling, chi-squared and correlation analysis, PCA, discretisation into four score bands, and aggregation. These steps were exploratory; only the target creation and the encoding described next feed the models.

C. Feature Preparation

The inputs are the five categorical features. Gender, race/ethnicity, lunch type and test preparation are one-hot encoded, with a single column kept for binary features. Parental education has a natural order and is ordinal-encoded from some high school to master's degree. This gives nine input columns, as summarised in Table II. The encoder sits inside a scikit-learn pipeline, so it is fitted only on training data in each split or fold.

D. Model Training

Two regressors are trained. Linear Regression uses default settings. Random Forest Regressor uses 300 trees, a minimum of 10 samples per leaf and random_state 42. Linear Regression gave the best results (Section V), so Fig. 2 shows its structure: the nine encoded inputs are weighted, summed with an intercept b, and mapped to the predicted average score. The notebook does not report the fitted weights.

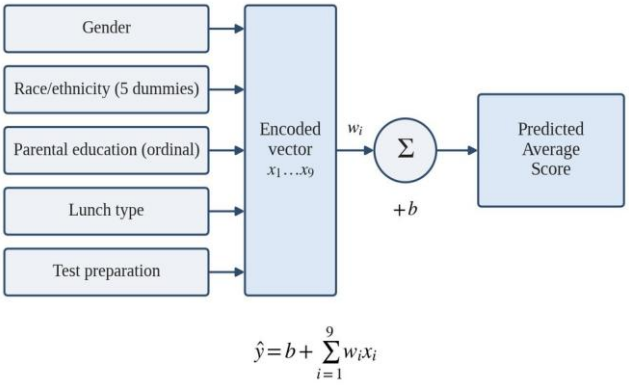


Fig. 2. Architecture of the best-performing model (Linear Regression).

E. Model Evaluation

Models are scored on the 20% test set with R^2, MAE and RMSE. Five-fold cross-validated R^2 on all 1,000 records gives a second estimate that does not depend on one split.

IV. EXPERIMENTAL SETUP

The work was run in Google Colab with Python, pandas, NumPy, matplotlib, seaborn and scikit-learn. Library versions, hardware, and training and prediction times were not reported. The target is Average_Score, the mean of the math, reading and writing scores. The data were split 80:20 (800 training and 200 test records) with random_state 42.
Table II summarises the dataset and processing steps.

TABLE II
DATASET AND PREPROCESSING SUMMARY

Feature Category	Number / Description	Processing Method
Nominal features	4: gender (2), race/ethnicity (5), lunch (2), test preparation (2)	One-hot encoding; one column kept for binary features
Ordinal feature	1: parental education (6 levels)	Ordinal encoding in natural order
Subject scores	3: math, reading, writing (0–100)	Averaged to form target; excluded from inputs
Target	Average_Score; mean 67.77	Mean of the three scores
Data quality	1,000 records; 0 missing; 0 duplicates; 211 distinct feature combinations	No cleaning needed
Score bands (exploratory)	Poor 30, Average 255, Good 517, Excellent 195*	Discretised with pd.cut; not used in models

Counts total 997 because the right-open bins exclude an average of exactly 100

The metrics are defined in (1)–(3) [15], where y_i is the actual score, $\hat{y}_i$ the predicted score, $\bar{y}$ the mean actual score and n the number of test records.

$$R^2 = 1 - \Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2 \quad (1)$$

$$MAE = (1/n)\ \Sigma\ |y_i - \hat{y}_i| \quad (2)$$

$$RMSE = \sqrt{[(1/n)\ \Sigma\ (y_i - \hat{y}_i)^2]} \quad (3)$$

MAE and RMSE are in score points. A higher R^2 and lower MAE and RMSE are better. Table III lists the full model configuration.

TABLE III
MODEL AND EVALUATION CONFIGURATION

Item	Setting
Train/test split	80:20 (800/200 records), random_state = 42
Nominal encoding	OneHotEncoder, drop = if_binary, handle_unknown = ignore
Ordinal encoding	OrdinalEncoder with fixed education order
Linear Regression	scikit-learn default settings
Random Forest	n_estimators = 300, min_samples_leaf = 10, random_state = 42
Validation	5-fold cross-validation of the full pipeline (R^2)
Reported metrics	R^2, MAE, RMSE (test); R^2 (cross-validation)

V. RESULTS AND DISCUSSION

A. Model Comparison

Table IV compares the two models. Linear Regression scored better on all four measures: a test R^2 of 0.178 against 0.127, an MAE of 10.39 against 10.67, an RMSE of 13.28 against 13.68, and a cross-validated R^2 of 0.215 against 0.165. Fig. 3 shows the same ordering. The gaps are small, about 0.05 in R^2 and under half a score point in MAE, so the two models are close in practice. The 1,000 records contain only 211 distinct combinations of the five features, so students with identical inputs can still score very differently. This sets a ceiling on the R^2 that any model built on these features can reach.

TABLE IV
PERFORMANCE COMPARISON OF MACHINE LEARNING MODELS

Model	Test R^2	MAE	RMSE	5-fold CV R^2
Linear Regression	**0.178**	**10.389**	**13.278**	**0.215**
Random Forest	0.127	10.674	13.679	0.165

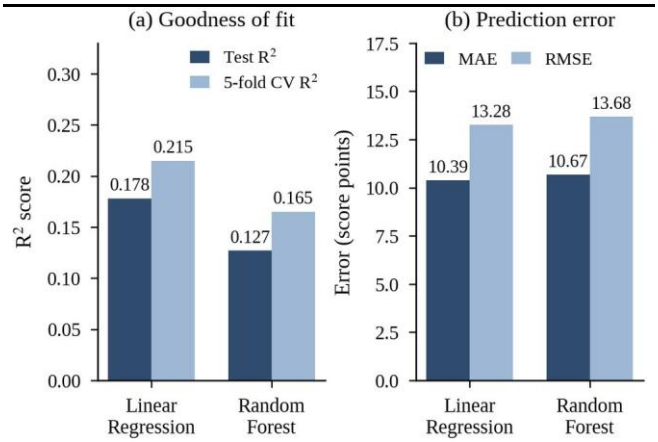


Fig. 3. Comparison of the two models: (a) R^2 on the test set and under fivefold cross-validation; (b) MAE and RMSE on the test set.

Both R^2 values are low. An R^2 of 0.178 leaves about 82% of the variance unexplained, and an RMSE near 13 points is close to the spread of the subject scores themselves (standard deviations of 14.6 to 15.2). Background features alone therefore give only a modest estimate of a student's average score. The cross-validated R^2 is higher than the test R^2 for both models, which suggests that the single 20% split was somewhat unfavourable and that estimates vary with the sample.

A possible reason for the linear model's edge is that the five categorical inputs act mostly additively on the score. A linear model can capture this with few parameters, while the

forest's extra flexibility brought no gain on this data. The notebook did not test this explanation.

B. Exploratory Findings

The notebook's exploratory steps give context for these results. Math, reading and writing scores correlate with Average_Score at 0.919, 0.970 and 0.966. A PCA on the standardised score columns found that the first component explains 94.4% of the variance, the second 4.7% and the third 0.9%. The last two components carry no variance because Total_Score and Average_Score are linear combinations of the three scores. Discretising Average_Score into four bands gave 30 Poor, 255 Average, 517 Good and 195 Excellent students, so most students fall in the middle bands. Grouping by the five categorical features reduced the 1,000 records to 211 groups, 106 female and 105 male.

C. Prediction Behaviour

Fig. 4 plots predicted against actual average scores for Random Forest on the test set, the only model plotted in the notebook. Predictions stay roughly between 50 and 82, although actual scores range from about 10 to 97. High scores are underestimated and low scores overestimated, which is expected when the inputs explain little variance. Only 30 of the 1,000 students fall in the lowest score band (Table II), so very low scores are the hardest to predict.

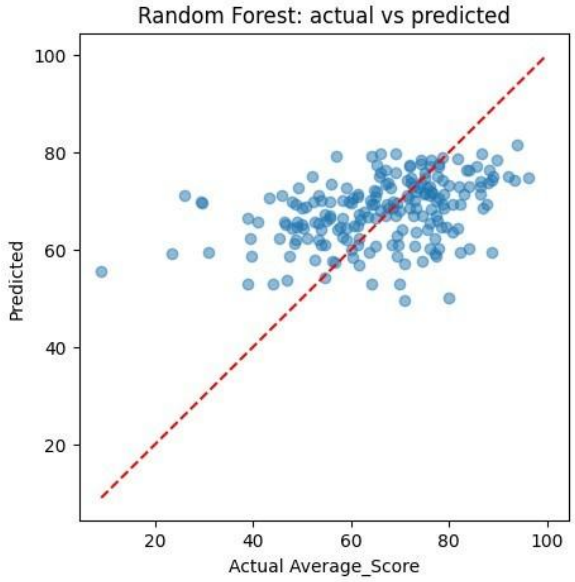


Fig. 4. Predicted versus actual Average_Score for Random Forest on the test set. The dashed line marks perfect prediction.

D. Feature Relevance

Table V lists chi-squared scores from the exploratory analysis. Lunch type ranks first, followed by test preparation and race/ethnicity. Fig. 5 shows the Random Forest importances. Lunch type, test preparation and parental education are the three largest single columns, at roughly 0.29, 0.24 and 0.22, and gender follows at about 0.10. The five race/ethnicity columns are each below 0.04 but add up to roughly 0.15 together, which agrees with its chi-squared rank. Lunch type may reflect household circumstances, but the dataset gives no further detail, so no causal reading is made.

TABLE V
CHI-SQUARED SCORES OF CATEGORICAL FEATURES (EXPLORATORY)

Feature	Encoded Columns	χ^2 Score
Lunch type	1	28.54
Test preparation course	1	20.86
Race/ethnicity	5	19.01
Parental level of education	1	11.43
Gender	1	7.53

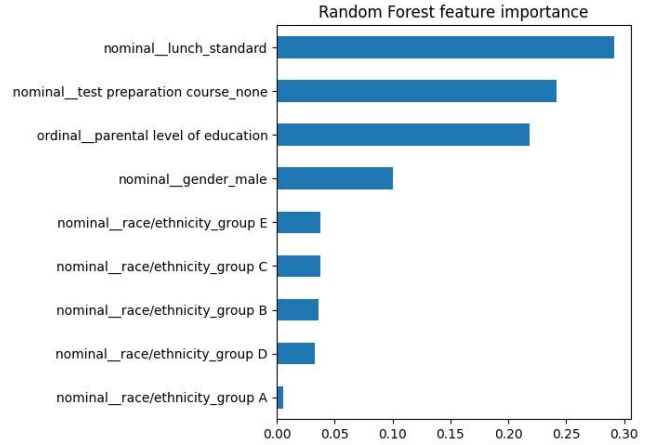


Fig. 5. Random Forest feature importance for the nine encoded input columns.

E. Comparison with Related Work

The R^2 here is much lower than the 0.8911 reported for GPA prediction from earlier scores [6], and below the 0.383 reported for a mixed academic and demographic feature set [10]. This is expected, because [6] uses earlier scores and [10] adds academic records, while this work uses background features only. The result that Linear Regression matches or beats Random Forest agrees with [6] and [7]. Direct comparison of values is not possible, since the datasets, targets and feature sets differ.

F. Limitations

This work has limits. The data are one public set of 1,000 records, and only five background features are used. Random Forest settings were fixed, not tuned, and one random seed was used for the split. Training and prediction times were not recorded, and the plots in Figs. 4 and 5 cover Random Forest only. The results describe this dataset and should not be taken as general.

VI. CONCLUSION

This work predicted the average of math, reading and writing scores for 1,000 students from five background features. The data had no missing or duplicate records. Nominal features were one-hot encoded, parental education was ordinal-encoded, and the score columns were excluded to avoid leakage. Linear Regression and Random Forest were compared on a held-out split and with five-fold crossvalidation. Linear Regression performed best, with a test R^2 of 0.178, an MAE of 10.39 and an RMSE of 13.28, although its margin over Random Forest was small. Lunch type, test preparation and parental education carried most of the information. The low R^2 shows that background features give

only a modest estimate of exam performance, so the results are best used as a baseline for richer models.

VII. FUTURE WORK

Future work will add features that describe study behaviour, such as attendance and earlier-term grades, since background features explained only about a fifth of the variance. Larger multi-school datasets and external validation are needed to test whether the findings hold elsewhere. The Random Forest settings should be tuned, and regularised linear models and gradient boosting added to the comparison. Explainability tools such as SHAP could give per-student explanations, and class balancing or targeted models could be tried for the small group of low scorers.

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